CONNECTED LIMITS IN A SLICE CATEGORY AGREE WITH LIMITS IN THE AMBIENT CATEGORY

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Editorial note. Counted span is the author's own proof environment, source lines 1897-1913, following the statement at 1884-1895.

Editorial note. The proof is complete with no omission markers.

Allowed standard prerequisites (editorial).

- Connected index categories and compatible families. the connectedness hypothesis of the statement, used at proof lines 1905-1907. Supplies that the structural morphisms f_i = s_i composed with phi_i to X are all equal, and nonemptiness gives at least one.
- Universal property of the limit in the slice category. the limit object L -> X of M in C/X, fixed at proof line 1898. Supplies the unique morphism phi: W -> L over X at proof lines 1911-1912.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let X be an object of $\mathcal{C}$. Let $M : \mathcal{I} \to \mathcal{C}/X$ be a diagram in the category of objects over X. If the index category $\mathcal{I}$ is connected and the limit of M exists in $\mathcal{C}/X$, then the limit of the composition $\mathcal{I} \to \mathcal{C}/X \to \mathcal{C}$ exists and is the same.*

Proof. Let $L \to X$ be an object representing the limit in $\mathcal{C}/X$. Consider the functor

$$W \longmapsto \lim_i \mathrm{Mor}_{\mathcal{C}}(W, M_i).$$

Let (φ_i) be an element of the set on the right. Since each M_i comes equipped with a morphism $s_i : M_i \to X$ we get morphisms $f_i = s_i \circ \varphi_i : W \to X$. But as $\mathcal{I}$ is connected we see that all f_i are equal. Since $\mathcal{I}$ is nonempty there is at least one f_i. Hence this common value $W \to X$ defines the structure of an object of W in $\mathcal{C}/X$ and (φ_i) defines an element of $\lim_i \mathrm{Mor}_{\mathcal{C}/X}(W, M_i)$. Thus we obtain a unique morphism $\phi : W \to L$ such that φ_i is the composition of ϕ with $L \to M_i$ as desired. □

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